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Extra Credit

Problem 3.1 Rajlich page 46

```
class Student:
    def __init__(self, student_id: int, grade: float = 0.0):
        if not isinstance(student_id, int):
            raise TypeError("Student ID must be an integer.")
        if not (0.0 <= grade <= 100.0):
            raise ValueError("Grade must be between 0.0 and 100.0.")
        self.student_id = student_id
        self.grade = grade

    def __repr__(self):
        return f"Student(id={self.student_id}, grade={self.grade:.2f})"

class Classroom:
    CLASS_SIZE = 30

    def __init__(self):
        # Initialize 30 students with IDs 1–30 and a default grade of 0.0
        self.students = [Student(student_id=i) for i in range(1, self.CLASS_SIZE + 1)]

    def _find_student(self, student_id: int) -> Student:
        for student in self.students:
            if student.student_id == student_id:
                return student
        raise ValueError(f"No student found with ID {student_id}.")

    def change_grade(self, student_id: int, new_grade: float) -> None:
        if not (0.0 <= new_grade <= 100.0):
            raise ValueError("Grade must be between 0.0 and 100.0.")
        student = self._find_student(student_id)
        student.grade = new_grade
        print(f"Updated: Student {student_id} → grade {new_grade:.2f}")
```

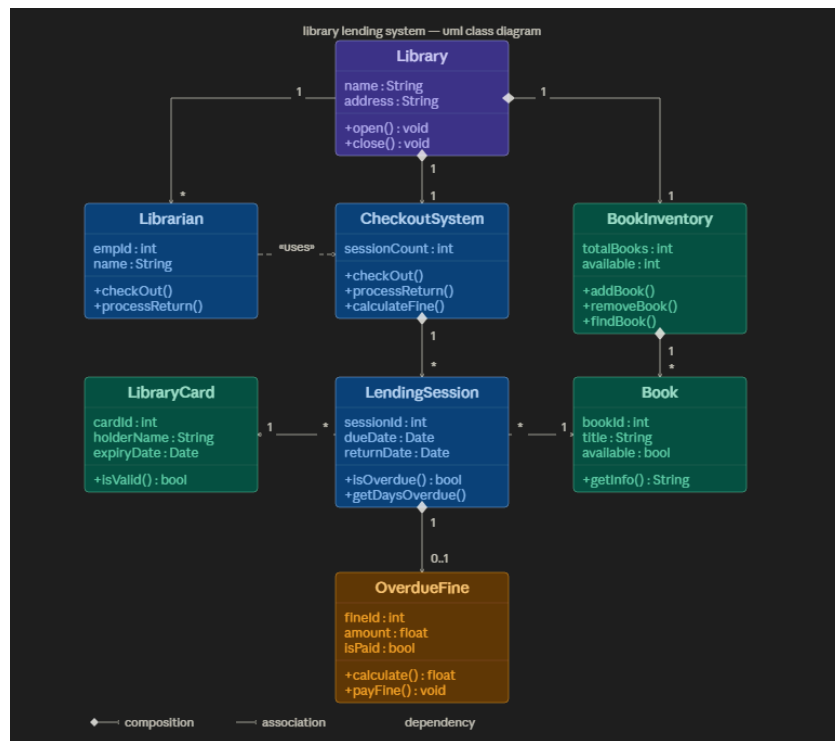
```

def compute_gpa(self) -> float:
    total = sum(student.grade for student in self.students)
    average = total / self.CLASS_SIZE
    return average

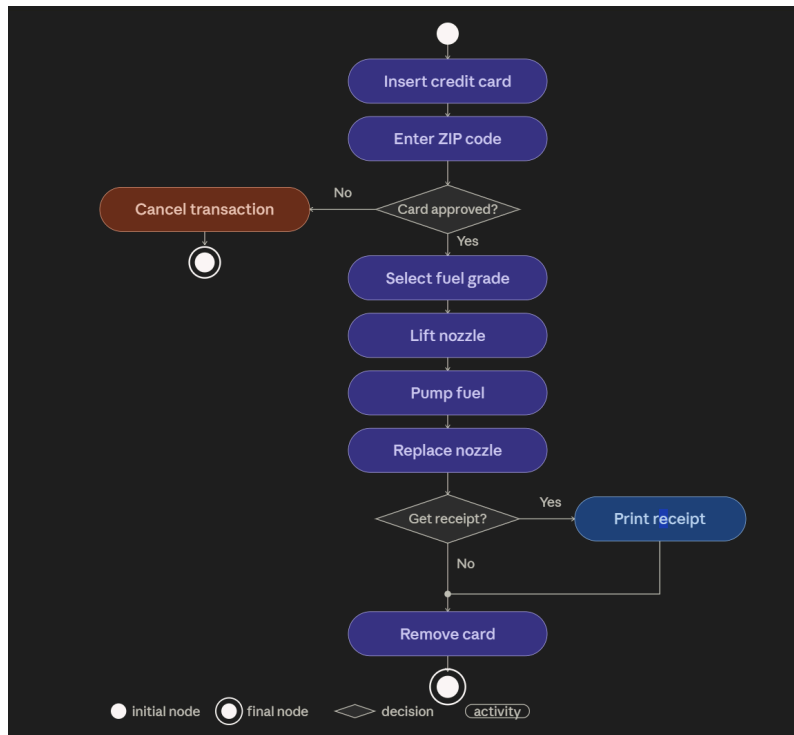
def display_roster(self) -> None:
    print(f'{'ID':>4} {'Grade':>6}')
    print("-" * 14)
    for student in self.students:
        print(f'{'student.student_id':>4} {'student.grade':>6.2f}')
    print("-" * 14)
    print(f'{'GPA':>4} {'self.compute_gpa():>6.2f'}\n")

```

Problem 4.2 Rajlich page 63



Problem 4.3, Rajlich page 63



Problem 10.6, Schach page 166

T1: $i = 10, j = 20 \Rightarrow i > 30 \rightarrow \text{false}, i == j \rightarrow \text{false}$
T2: $i = 15, j = 20 \Rightarrow i > 30 \rightarrow \text{false}, i == j \ \&\& \ i \leq 30 \rightarrow \text{true}$
T3: $i = 40, j = 100 \Rightarrow i > 30 \rightarrow \text{true}, i \leq 60 \ \&\& \ j \leq 150 \rightarrow \text{true}$
T4: $i = 70, j = 100 \Rightarrow i > 30 \rightarrow \text{true}, i \leq 60 \rightarrow \text{false} \text{ else if } i \leq 90 \rightarrow \text{false}$
T5: $i = 100, j = 200 \Rightarrow i > 30 \rightarrow \text{true}, i \leq 60 \rightarrow \text{false}, i \leq 90 \rightarrow \text{false}$

Problem 10.8, Schach page 167

```
assert( testGrade.getFinalGrade(70).equals("Pass") );
assert( testGrade.getFinalGrade(69).equals("Fail") );
```

Problem 10.8, Scratch page 167

// Testing Grade A

```
assert( testGrade.getFinalGrade(100).equals("A") );  
assert( testGrade.getFinalGrade(90).equals("A") );
```

// Testing Grade B

```
assert( testGrade.getFinalGrade(89).equals("B") );  
assert( testGrade.getFinalGrade(80).equals("B") );
```

// Testing Grade C

```
assert( testGrade.getFinalGrade(79).equals("C") );  
assert( testGrade.getFinalGrade(70).equals("C") );
```

// Testing Grade D

```
assert( testGrade.getFinalGrade(69).equals("D") );  
assert( testGrade.getFinalGrade(60).equals("D") );
```

// Testing Grade F

```
assert( testGrade.getFinalGrade(59).equals("F") );  
assert( testGrade.getFinalGrade(0).equals("F") );
```

Problem 11.2, Schach page 388

Ambiguities

- Mushroom and sugar are used but aren't in the ingredients list
- Frozen orange juice is asked for but never specified when to defrost it

Contradiction

- Ingredients only specify one lemon
- There is no mention of mushrooms

Ambiguity

- Is a can of orange juice considered fresh?
- Where do we get the lemons plus not added in ingredients
- Pour into what?